LELEC2870 | MACHINE LEARNING PROJECT | 2026-2027

Part 1 - Linear model (baseline)

Smurf heart-failure risk | Interim report - 3 November 2026

Objective and dataset

We predict the 10-year heart-failure risk from tabular medical data. The labeled dataset contains 1,000 training and 500 held-out test patients. Images were deliberately excluded from this first linear baseline.

Preprocessing

We removed the heart-scan filename/identifiers and retained only clinical and lifestyle variables. Missing numerical values were imputed by the training-fold median and then standardized. Categorical values (profession and consumption levels) were imputed with the most frequent training-fold category and one-hot encoded; unknown categories were ignored at prediction time. No missing values were found in the provided data; imputation was kept for robustness. All transformations were fitted inside each training fold to prevent leakage.

Feature selection

Forward wrapper selection minimized mean five-fold cross-validated RW-RMSE on the 1,000 training cases, using risk-weighted Ridge regression as the selection model. The number of *original* variables was capped at six (one-hot columns of one variable count as one). **Selected (6): blood pressure, hemoglobin, weight, height, age, sarsaparilla.** The CV RW-RMSE decreased from 0.0662 with one feature to 0.0600 with six.

Model selection and assessment

We compared ordinary least squares with Ridge regression (alpha = 0.01 to 1,000), weighted versus unweighted training, and optional [0,1] output clipping, using the same five CV folds and the guideline-defined RW-RMSE. The selected model was **Weighted least squares**, risk-weighted fitting, clipped to [0, 1]. The test set was evaluated only after these choices were fixed.

Results

Evaluation set	RW-RMSE	RMSE
5-fold validation (mean +/- SD)	0.0600 +/- 0.0045	Not computed
Training (1,000 patients)	0.0588	0.0530
Held-out test (500 patients)	**0.0646**	0.0554

Interpretation and limitations

On the held-out test, RW-RMSE was 0.0646, compared with 0.1025 for a weighted constant-risk baseline (37.0% lower). As feature selection and tuning both used the training CV folds, the held-out test is the more reliable generalization estimate. Selected features are predictive indicators, not proof of causality. For the 500 submitted predictions, the selected configuration was refitted on all 1,500 labeled cases.

Sources: LELEC2870 course slides (Linear Regression; Feature Selection; Model Selection); LELEC2870 Project Guidelines 2026-2027. Software: Python, scikit-learn 1.8.0.